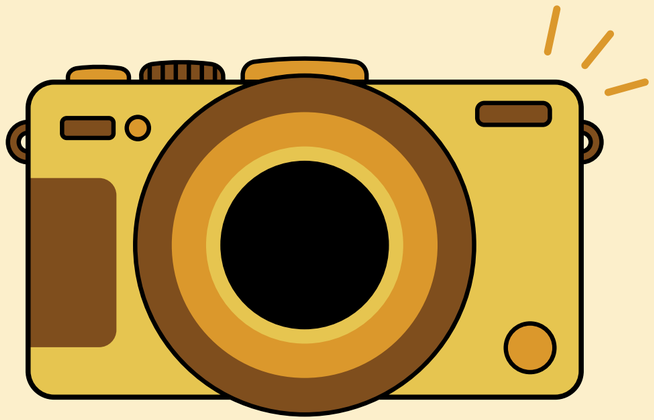

AIDriven Archaeological Site Mapping

AIML INTERNSHIP-2025

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INTRODUCTION

“How much of our history remains undiscovered simply because it lies hidden beneath terrain, vegetation, or centuries of erosion?”

Archaeologists and researchers often rely on manual inspection of satellite and drone imagery, a tedious and error-prone process. Our project proposes an AI-driven mapping platform that automates this discovery — identifying ancient structures, segmenting terrain, and predicting erosion-prone zones with high precision.

PROBLEM STATEMENT AND SOLUTION

PROBLEM STATEMENT

Building an AI-based platform that analyzes satellite and drone imagery to assist archaeologists in segmenting ancient ruins and vegetation, detecting and classifying artifact structures, and predicting erosion-prone zones. This system will support conservation planning and archaeological research.

SOLUTION

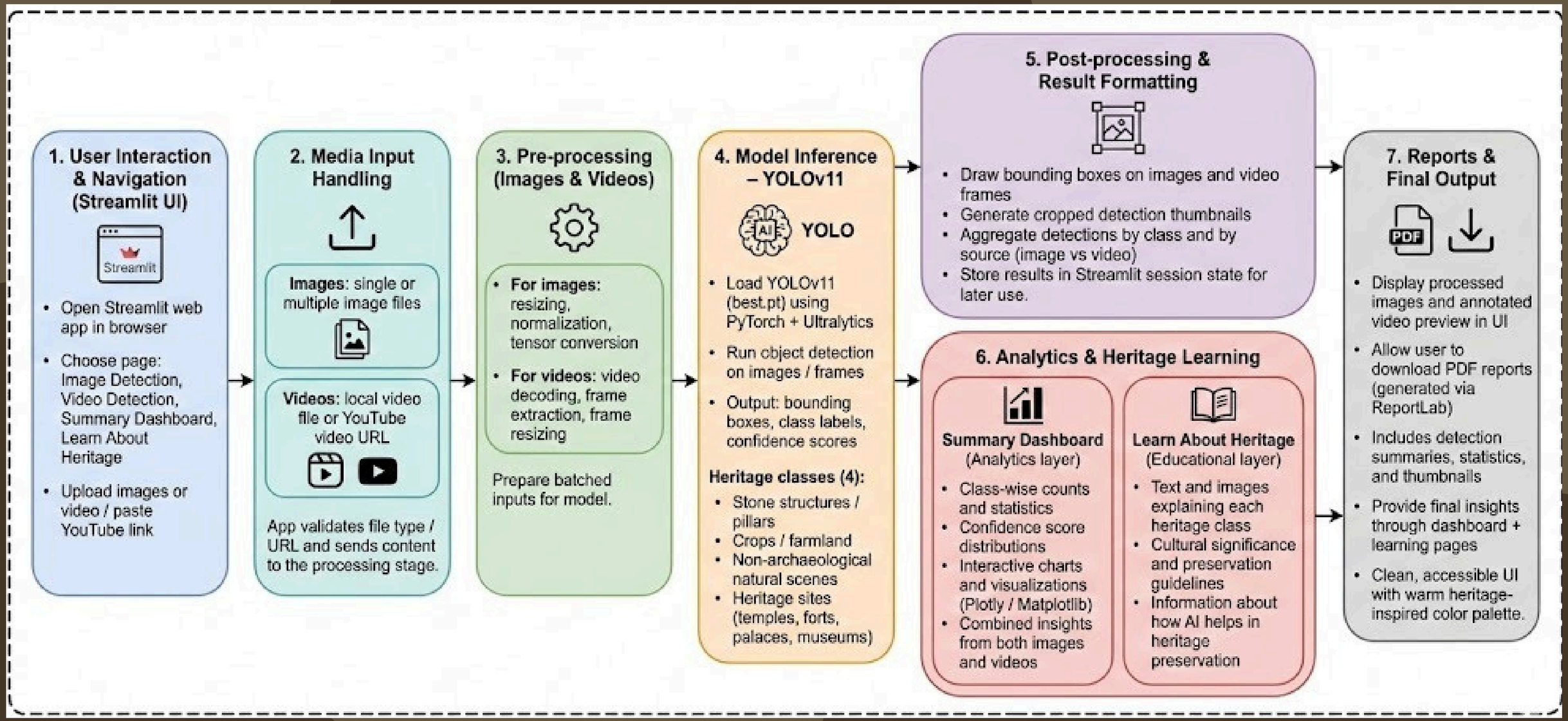
We built an AI-powered imagery analysis platform that:

- Performs semantic segmentation to separate ruins, vegetation, and terrain.
- Uses object detection to identify and classify archaeological features.
- Predicts erosion-prone zones using ML models trained on terrain patterns.
- Displays outputs through an interactive geospatial dashboard, enabling researchers to explore, filter, and export insights for field investigation.

MOTIVATION

- **Accelerating Scientific Discovery:** Automating imagery analysis reduces months of manual effort and enables researchers to work at scale.
- **High Economic Value:** Early detection of erosion and site degradation prevents costly restoration, preserves cultural tourism revenue, and reduces field survey expenses.
- **Growing Industry Demand:** Geospatial AI is rapidly expanding across environmental monitoring, agriculture, mining, and defense — making such a system commercially viable beyond archaeology.
- **Bridging Research and Real-World Application**: Our platform shows how AI models can convert raw imagery into actionable insights, empowering both academic researchers and industry teams.

WORKFLOW


1. User Interaction & Navigation (Streamlit UI)
Streamlit
• Open Streamlit web app in browser
• Choose page: Image Detection, Video Detection, Summary Dashboard, Learn About Heritage
• Upload images or video / paste YouTube link
2. Media Input Handling
Images: single or multiple image files
Videos: local video file or YouTube video URL
App validates file type / URL and sends content to the processing stage.
3. Pre-processing (Images & Videos)
• For images: resizing, normalization, tensor conversion
• For videos: video decoding, frame extraction, frame resizing
Prepare batched inputs for model.
4. Model Inference – YOLOv11
YOLO
• Load YOLOv11 (best.pt) using PyTorch + Ultralytics
• Run object detection on images / frames
• Output: bounding boxes, class labels, confidence scores
Heritage classes (4):
• Stone structures / pillars
• Crops / farmland
• Non-archaeological natural scenes
• Heritage sites (temples, forts, palaces, museums)
5. Post-processing & Result Formatting
• Draw bounding boxes on images and video frames
• Generate cropped detection thumbnails
• Aggregate detections by class and by source (image vs video)
• Store results in Streamlit session state for later use.
6. Analytics & Heritage Learning
Summary Dashboard (Analytics layer)
• Class-wise counts and statistics
• Confidence score distributions
• Interactive charts and visualizations (Plotly / Matplotlib)
• Combined insights from both images and videos
Learn About Heritage (Educational layer)
• Text and images explaining each heritage class
• Cultural significance and preservation guidelines
• Information about how AI helps in heritage preservation
7. Reports & Final Output
PDF
• Display processed images and annotated video preview in UI
• Allow user to download PDF reports (generated via ReportLab)
• Includes detection summaries, statistics, and thumbnails
• Provide final insights through dashboard + learning pages
• Clean, accessible UI with warm heritage-inspired color palette.

DATASET PREPERATION

Dataset size:

- **Images: 6488**

Sources:

- **[Google Earth Pro] (https://www.google.com/earth/versions/earthpro)**
- **[OpenAerialMap] (https://openaerialmap.org)**
- **Sentinel-2 Satellite Imagery (ESA – European Space Agency)**
- **Drone Imagery (DJI, Parrot) – Field Survey**
- **Custom annotated images (manually labeled or via QGIS/Labelbox)**

Split:

- **train..... 4524(70%)**
- **test....1292(20%)**
- **val....651(10%)**

MODEL TRAINING AND TESTING

Model Training Overview

- Used YOLOv11 for object detection & U-Net/DeepLab for segmentation.
- Models optimized for satellite + drone imagery used in archaeology.

Training Pipeline

- Dataset split: 70% Train | 20% Val | 10% Test
- Preprocessing: tiling, resizing, normalization.
- Training setup:
  - Batch size: 16
  - Epochs: 50–100
  - Losses: YOLO → Objectness/Classification/Localization
  - U-Net → Dice + Cross-Entropy
- GPU acceleration via Google Colab.

TOOLS AND FRAMEWORKS USED

- **Programming Language: Python – Used for model development, training, and data processing**
- **Frameworks: PyTorch, Ultralytics YOLOv11 – For deep learning and object detection**
- **Libraries: NumPy, Pandas, OpenCV, Matplotlib – For data handling, image processing, and visualization**
- **Annotation Tool: LabelImg – Used for labeling archaeological objects and defining bounding boxes**
- **Hardware: NVIDIA RTX 4050 GPU – For model training and real-time inference**

RESULTS

Model Performance

- High detection accuracy across four classes using YOLOv11.
- Strong detection of heritage sites and stone structures even in noisy backgrounds.
- Reliable classification of crop/farmland and non-archaeological regions.
- Segmentation models accurately highlighted ruins, terrain lines, and feature boundaries.

Real-Time Detection

- Image mode: Precise bounding boxes with >85% confidence for most classes.
- Video mode: Smooth frame-by-frame detection at low and high resolutions.
- Stable performance across satellite, drone, and ground imagery.

Dashboard Insights

- Auto-generated charts: class counts, confidence distribution, detection summaries.
- Exportable PDF reports with annotated outputs and metrics.
- Clean interface for browsing images, videos, statistics, and erosion predictions.

APP RESULTS-IMAGE DETECTION

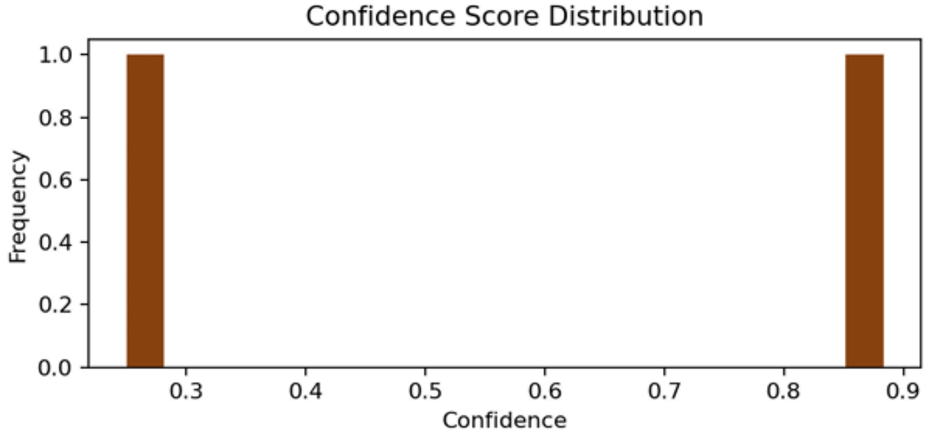
Executive Summary

In the analyzed content, 2 heritage objects were detected with an average confidence of 56.7%. The most common detection was Heritage Sites (temples, palaces, forts, museums) (100% of all detections).

Detection Statistics

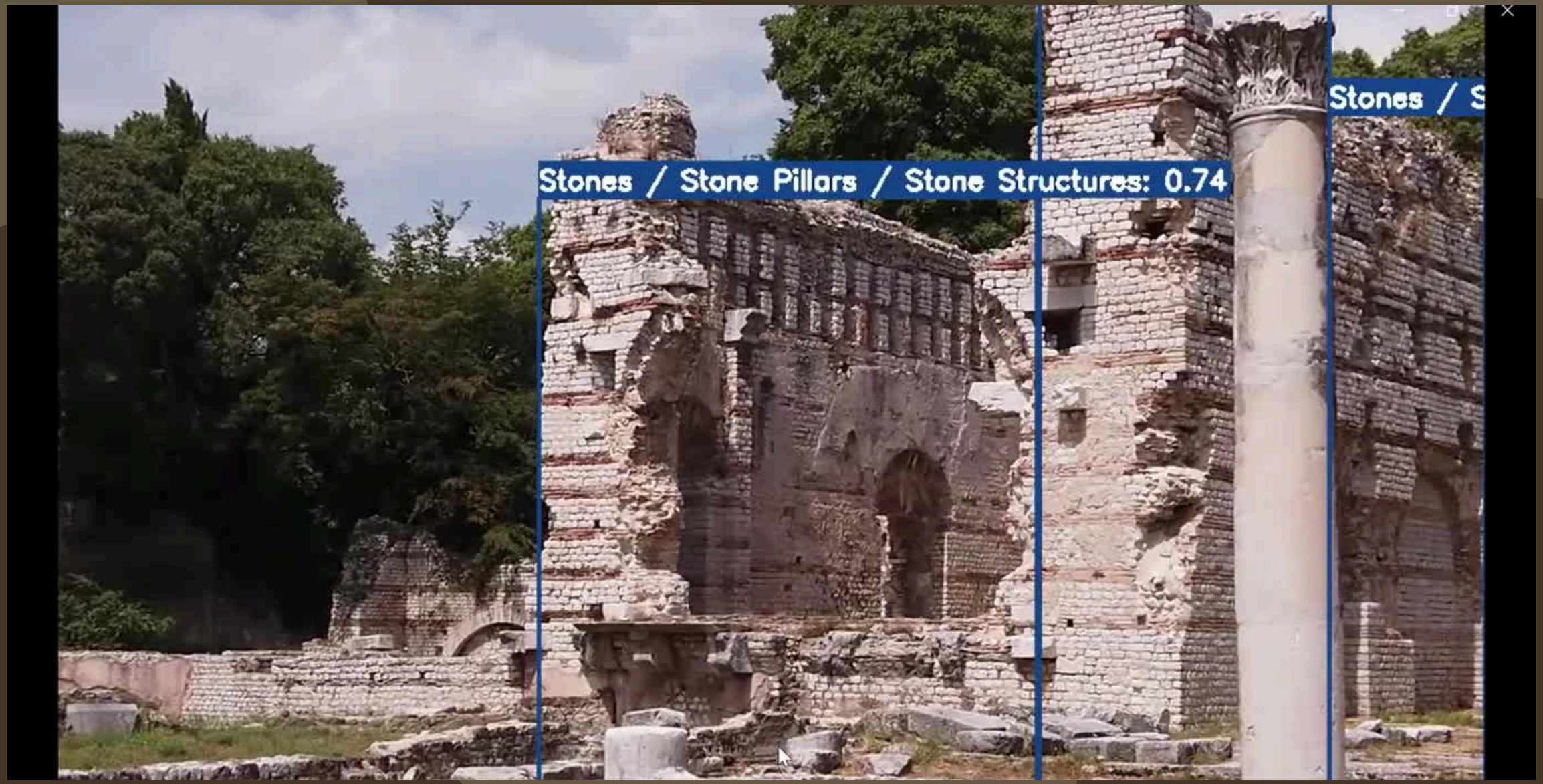
Metric	Value
Total Detections	2
Average Confidence	56.7%
Heritage Sites (temples, palaces, forts, museums) Count	2

Visualizations



Sample Detections

APP RESULTS-VIDEO DETECTION



APP RESULTS-VIDEO DETECTION

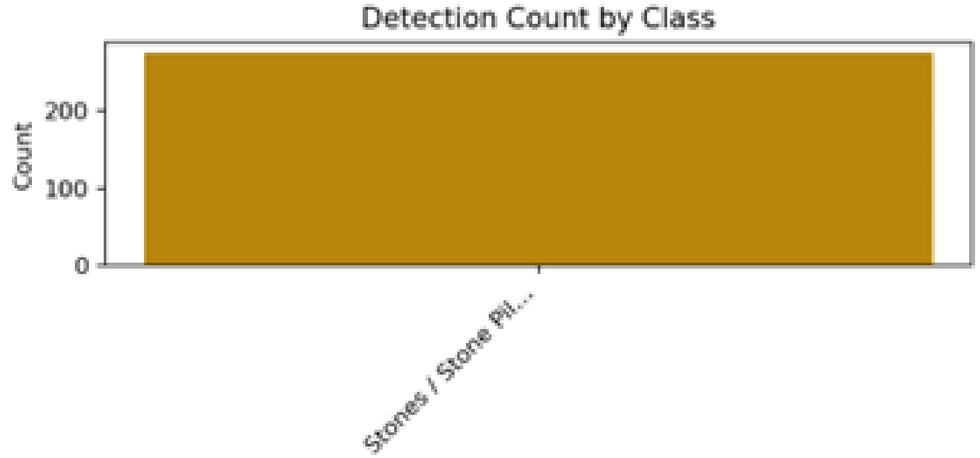
Executive Summary

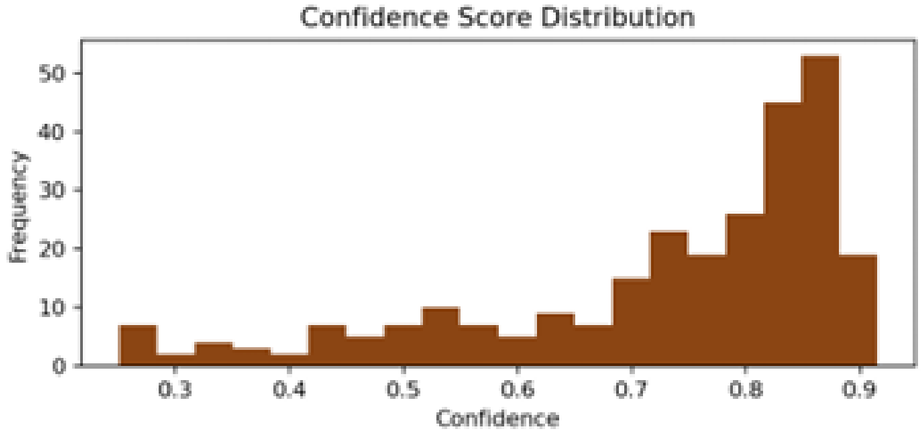
Video Analysis Report Duration: 16.0 seconds In the analyzed content (0m 16s of footage), 275 heritage objects were detected with an average confidence of 72.8%. The most common detection was Stones / Stone Pillars / Stone Structures (100% of all detections).

Detection Statistics

Metric	Value
Total Detections	275
Average Confidence	72.8%
Stones / Stone Pillars / Stone Structures Count	275

Visualizations





Sample Detections

FUTURE SCOPE

Enhancing Heritage Detection and Preservation

We're planning to enhance our heritage detection model with finer-grained classification, including architectural styles like Dravidian, Indo-Islamic, and Colonial, as well as region-specific heritage types.

Real-time Detection & Geo-tagging

Future updates will include real-time camera detection, live webcam feed support, and geo-tagging with GPS metadata for automatic site mapping and heritage inventories.

Mobile & Edge Deployment

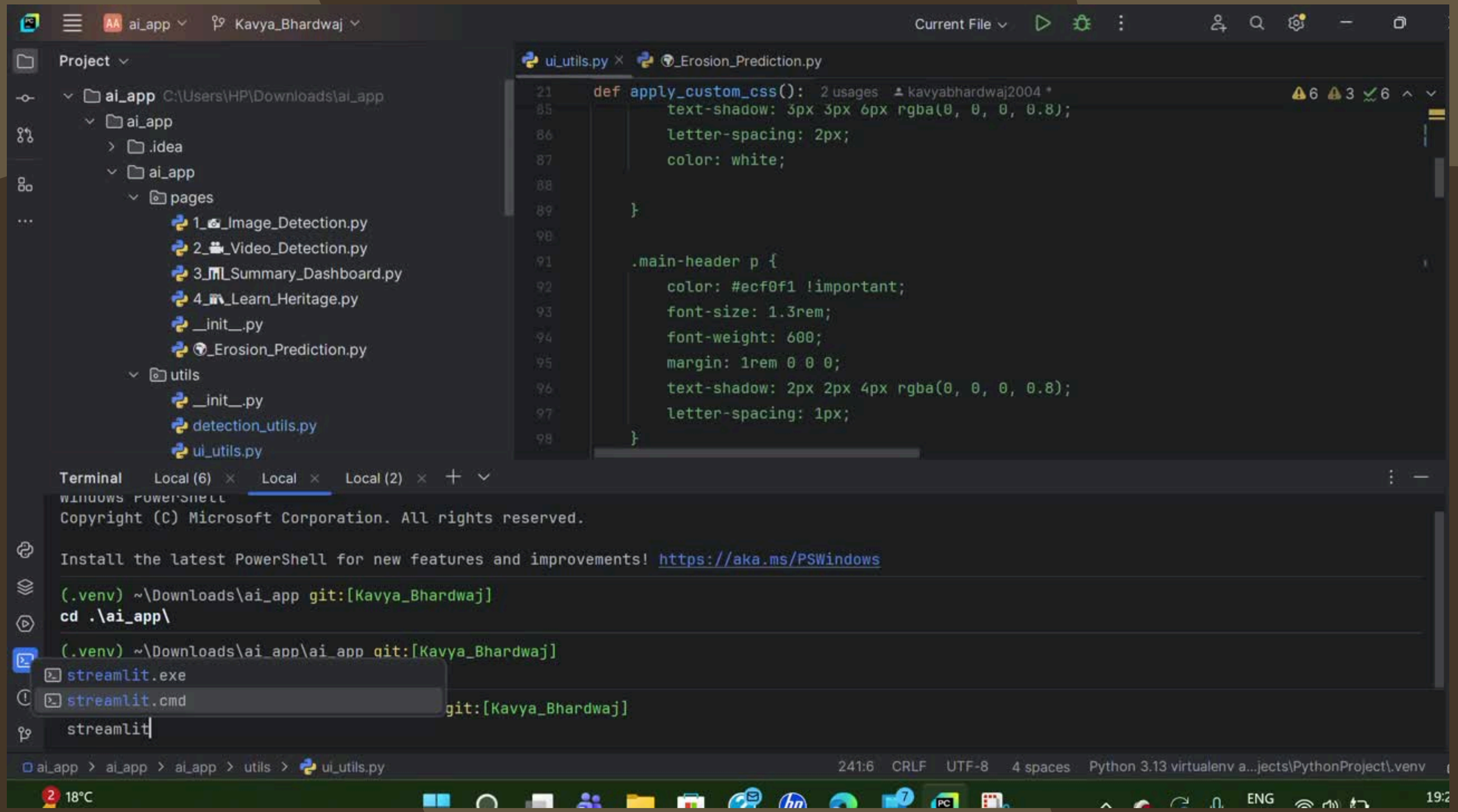
We'll be converting our model to ONNX/TFLite format, enabling deployment on mobiles and tablets for field surveys and archaeology teams.

Integration & Expansion

We're looking at integrating with AR/VR tech, overlaying heritage info on live camera views and 3D visualizations, and deploying a cloud-based platform for multi-user access and centralized data storage.

- HeritageLens AI demonstrates how modern computer vision can be effectively used to support cultural preservation.
- By integrating the YOLOv11 model with a clean Streamlit interface, the system allows users to easily upload images or videos, perform heritage detection, view annotated outputs, and access meaningful insights through dashboards and learning modules.
- The project successfully delivers a user-friendly platform that simplifies heritage identification and encourages broader understanding and appreciation of cultural sites.
- It highlights the potential of AI as a valuable tool for researchers, students, and heritage organizations.

POC VIDEO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Kavya_Bhardwaj
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.idea
ai_app
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1_Image_Detection.py
2_Video_Detection.py
3_Summary_Dashboard.py
4_Learn_Heritage.py
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_Erosion_Prediction.py
utils
__init__.py
detection_utils.py
ui_utils.py
ui_utils.py
_Erosion_Prediction.py
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letter-spacing: 2px;
color: white;
}
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letter-spacing: 1px;
}
Terminal
Local (6)
Local
Local (2)
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows
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Thank you so much